

A Nonlinear Kapitza Law with Multiple Steady States

**Exact construction, bifurcation structure, and numerical
verification**

Discussion draft

July 21, 2026

The maximum principle bounds the temperature field

Full transmission problem

The maximum principle gives

$$\inf_{\Gamma} g \leq u(x) \leq \sup_{\Gamma} g.$$

Hence normalized boundary data give $0 \leq u \leq 1$. This bounds u , but does not determine the sign of $[u]$ by itself.

One-dimensional reduction

$$s = [u] = 1 - (1 + \delta)a,$$

$$\delta a = \beta(s)s.$$

For $\delta > 0$ and $\beta(s) > 0$, these relations imply

$$0 < a < \frac{1}{1 + \delta}, \quad 0 < s < 1.$$

Multiplicity is a fixed-point question

Eliminating the jump s gives

$$a = F(a) := \frac{\beta(1 - (1 + \delta)a)}{\delta + (1 + \delta)\beta(1 - (1 + \delta)a)}.$$

Geometric interpretation

Each steady state is an intersection of

$$y = F(a) \quad \text{and} \quad y = a.$$

Construction strategy

Choose a map with three fixed points first, then reconstruct the Kapitza law that produces it.

A sigmoid creates three steady states

Prescribe the fixed-point map

$$f_{\alpha}(a) = \frac{\gamma}{2} \left[1 + \tanh \left(\frac{\alpha}{2} \left(\frac{2a}{\gamma} - 1 \right) \right) \right], \quad \gamma = \frac{1}{1 + \delta}.$$

- ▶ $f_{\alpha}(\gamma/2) = \gamma/2$.
- ▶ The central fixed point is always present.
- ▶ Its slope is

$$f'_{\alpha}(\gamma/2) = \frac{\alpha}{2}.$$

Expected transition

Once the central slope exceeds 1, two additional intersections appear on either side.

The inverse construction gives an exponential law

Imposing $F = f_\alpha$ gives

$$\beta = \frac{\delta f_\alpha}{1 - (1 + \delta)f_\alpha}.$$

Since $a = (1 - s)/(1 + \delta)$,

$$\beta_0(s) = \frac{\delta}{1 + \delta} \exp [\alpha(1 - 2s)].$$

Positive

$$\beta_0(s) > 0$$

Decreasing

$$\beta'_0(s) < 0$$

Convex

$$\beta''_0(s) > 0$$

The threshold and all branches are explicit

Introduce

$$x = \frac{2a}{\gamma} - 1, \quad \kappa = \frac{\alpha}{2}.$$

The fixed-point equation becomes

$$x = \tanh(\kappa x).$$

Critical value

$$\kappa_c = 1, \quad \alpha_c = 2.$$

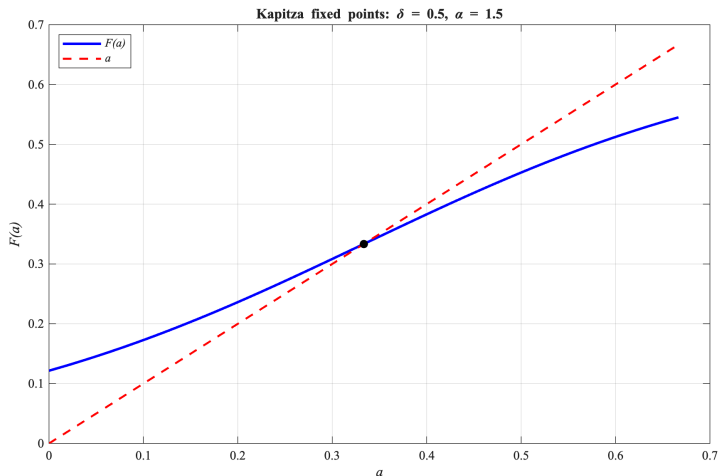
One solution below the threshold;
exactly three above it.

Outer branches

$$\alpha(x) = 2 \frac{\operatorname{artanh} x}{x}, \quad 0 < |x| < 1.$$

This is a supercritical pitchfork.

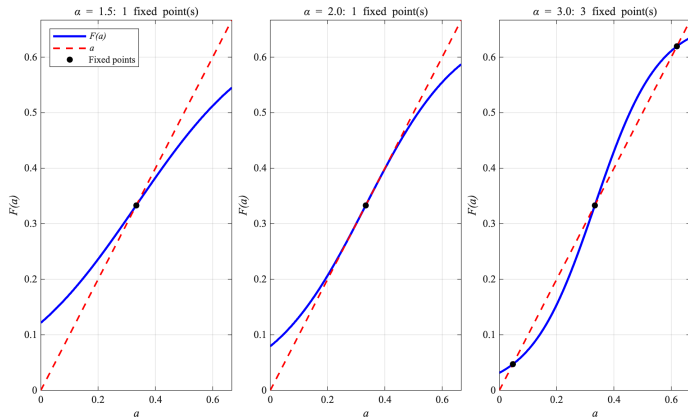
The teacher's example has one fixed point



For $\delta = 0.5$ and $\alpha = 1.5 < 2$, MATLAB gives $a = 1/3$, $s = 1/2$, and $q = 1/6$.

Three cases reveal the transition

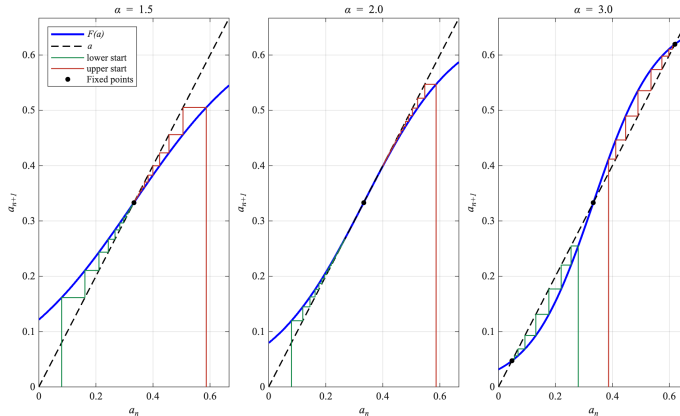
Fixed-point geometry across the pitchfork threshold



The critical map is tangent to $y = a$ at $a = \gamma/2$; two additional intersections appear when $\alpha > 2$.

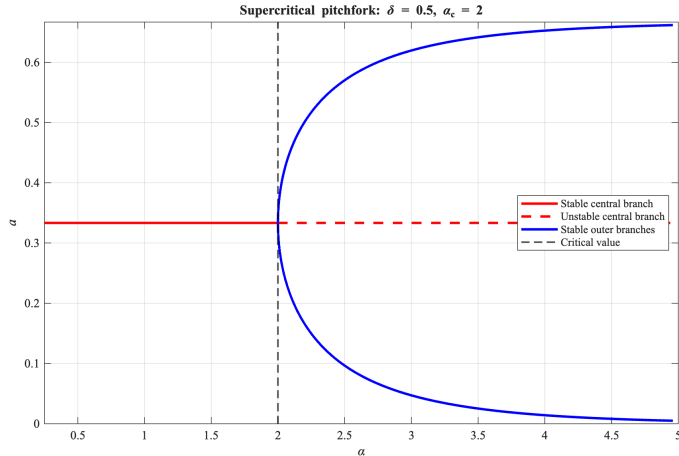
Cobwebs reveal stability and branch selection

Cobweb iterations reveal stability and branch selection



At $\alpha = 2$ convergence is slow; at $\alpha = 3$ two starts near the repelling center reveal selection of the attracting outer branches.

MATLAB recovers the supercritical pitchfork



Solid curves are attracting for fixed-point iteration; the dashed central segment is repelling for $\alpha > 2$.

The bias h has a direct physical meaning

Add a bias to the prescribed map:

$$f_{\alpha,h}(a) = \frac{\gamma}{2} \left[1 + \tanh \left(\frac{\alpha}{2} \left(\frac{2a}{\gamma} - 1 \right) + h \right) \right].$$

The reconstructed law is

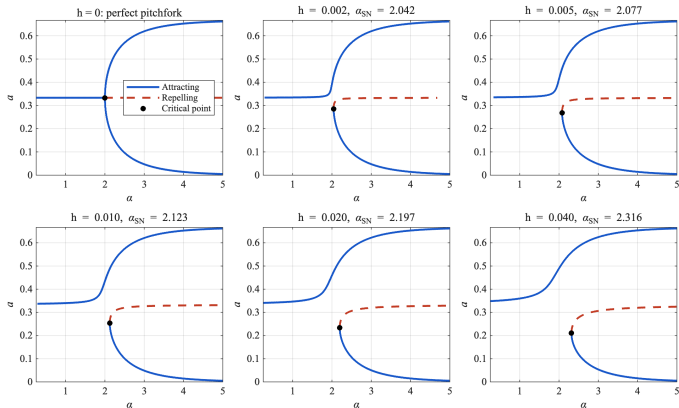
$$\beta_h(s) = \frac{\delta}{1 + \delta} \exp [\alpha(1 - 2s) + 2h] = e^{2h} \beta_0(s).$$

Physical meaning of h

h changes the overall amplitude of the Kapitza conductance law. Equivalently, $2h$ is the logarithmic change relative to the reference interface law.

Six small biases reveal the imperfect pitchfork

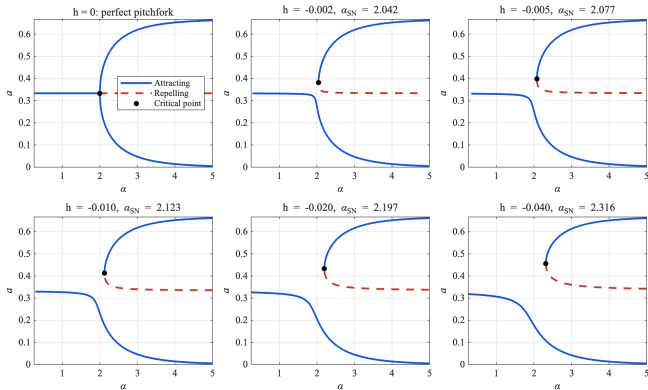
Progressive unfolding at six small fixed biases ($\delta = 0.5$; α is the control parameter)



With α as control parameter, every $h > 0$ breaks the exact pitchfork; the saddle node shifts according to $\alpha_{SN} - 2 \sim 2(3h/2)^{2/3}$.

Negative h gives the reflected unfolding

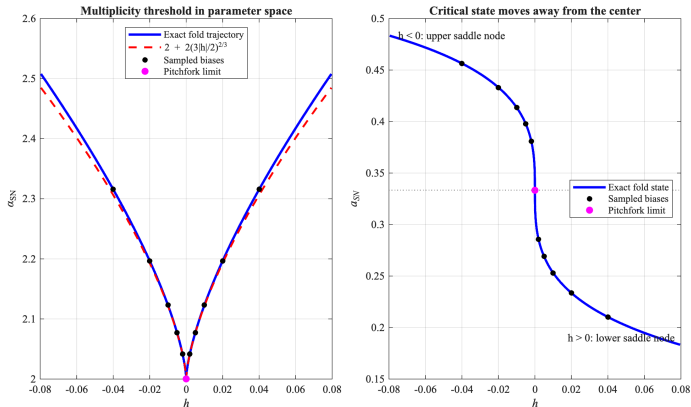
Mirror unfolding at six small nonpositive biases ($\delta = 0.5$; α is the control parameter)



By $(x, h) \mapsto (-x, -h)$, $a(\alpha; -h) = \gamma - a(\alpha; h)$ and $\alpha_{SN}(-h) = \alpha_{SN}(h)$: the branches exchange sides, but the threshold is unchanged.

The critical point follows a cusp trajectory

Saddle-node trajectory from the pitchfork limit ($\delta = 0.5$)



The pitchfork point $(h, \alpha, a) = (0, 2, 1/3)$ is the cusp tip. For $h \neq 0$, it continues as a saddle-node boundary.

The $h^{2/3}$ law follows from cancellation

At a saddle node,

$$h = \operatorname{artanh}(x) - \frac{x}{1-x^2}, \quad \alpha_{\text{SN}} = \frac{2}{1-x^2}.$$

Near $x = 0$,

$$\begin{aligned} \operatorname{artanh}(x) &= x + \frac{x^3}{3} + \frac{x^5}{5} + O(x^7), \\ \frac{x}{1-x^2} &= x + x^3 + x^5 + O(x^7). \end{aligned}$$

The linear terms cancel:

$$h = -\frac{2}{3}x^3 + O(x^5).$$

Thus, for $h > 0$,

$$x_{\text{SN}} \sim -\left(\frac{3h}{2}\right)^{1/3}.$$

Since $\alpha_{\text{SN}} - 2 \sim 2x_{\text{SN}}^2$,

$$\alpha_{\text{SN}} - 2 \sim 2\left(\frac{3|h|}{2}\right)^{2/3}.$$

Two folds bound the three-solution region

The reduced equation is

$$x = \tanh(\kappa x + h), \quad h(x) = \operatorname{artanh}(x) - \kappa x.$$

The fold condition $dh/dx = 0$ gives

$$x_f = \sqrt{1 - \frac{1}{\kappa}}, \quad h_c = \kappa x_f - \operatorname{artanh}(x_f) > 0.$$

Fold points

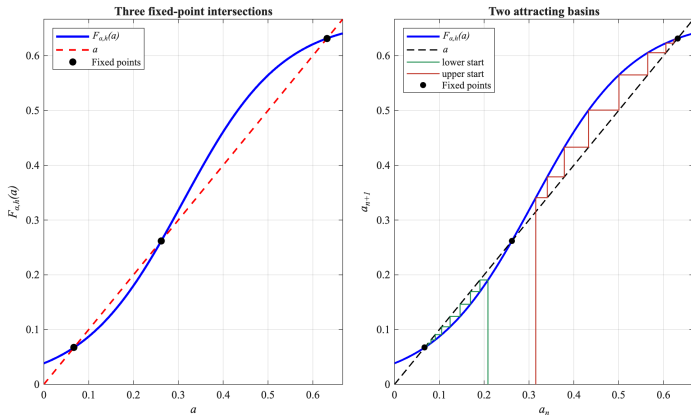
$$(h, x) = (h_c, -x_f), \quad (-h_c, x_f).$$

Number of solutions

$$\begin{array}{c|c} |h| < h_c & 3 \\ |h| = h_c & 2 \text{ distinct roots} \\ |h| > h_c & 1 \end{array}$$

The biased map has two attracting basins

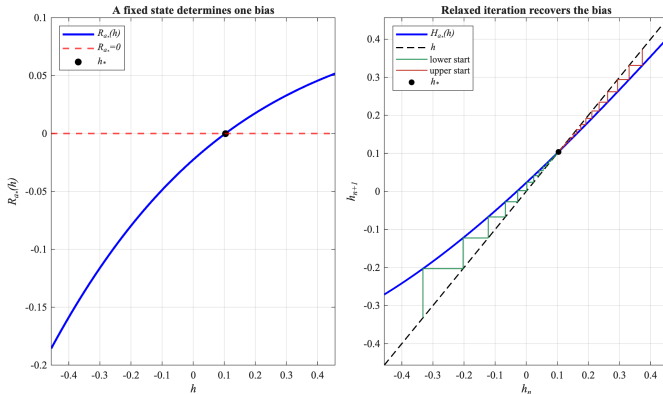
Fixed-h slice: $\delta = 0.5$, $\alpha = 3.0$, $h = 0.1038$



Here h is fixed and a_n is the iterated state. The starts lie on opposite sides of the repelling middle fixed point, exposing both basins.

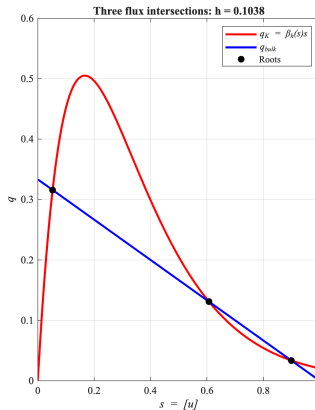
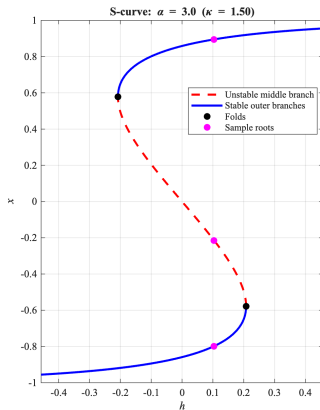
Fixing the state recovers one compatible bias

Fixed-a parameter recovery: $\delta = 0.5$, $\alpha = 3.0$, $a_* = 0.631645$, $h_* = 0.103773$



This h -cobweb visualizes numerical parameter inversion for a prescribed a_* . It is not the physical dynamics or the bifurcation diagram.

The S-curve has three flux intersections



For $\delta = 0.5$ and $\alpha = 3$: $\kappa = 1.5$, $x_f = 0.577350$, $h_c = 0.207546$.

Thank you

Questions and discussion